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(1) Research Program/Project/Study Title: Acceptability of Ready to Eat Darag Native Chicken and Bamboo Shoot Food Products	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Dumarao
Province:		Capiz
Region:		6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus)

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☐ Applied ☐ Tech. Promotion/Commercialization		☐ Collaborative Research (external agency)	
(7) Sector/Commodity Livestock and Crop		(8) Discipline/Field of Specialization Foods	
(8) Rationale Ready-to-eat, instant food mix, or heat-and-serve food are the most convenient food products present in the market today. It helps the majority of consumers in saving their time in cooking the healthy options foods. These ready-to-eat meals also will help them be practical by choosing healthy meals without spending several hours in the market and in the kitchen. Nowadays, these readily accessible foods are now growing in the market and placed in the market for retail or single-dose delivery. Over the past years, the importance of ready meals in the food markets has been growing steadily (Olsen <i>et al.</i>, 2010). In our modern world, the term “quick” can describe the way we are living. The propensity to move towards a faster lifestyle increases the demand for convenience in meal preparation (Costa <i>et al.</i>, 2007). With their busy lifestyles, consumers require products and services to ease and support their hectic lives (Olsen <i>et al.</i>, 2010). As a response to this demand, the food industry has focused on developing ready-to-eat (RTE meals) as a solution (Aloudat <i>et al.</i>, 2021). Despite of the growing market of this ready to eat meals, there are some disadvantages however that may occur, such as not healthy meals that may cause some diseases and or metabolic disorders to consumers. A healthy option ready to eat food product is, therefore the best option that may help consumers to enjoy their foods in a healthy way. One option that may be used for this RTE food product would be using the Darag native chicken and Bamboo shoots. Darag Native chicken, scientifically known as <i>Gallus gallus domesticus</i>, is a well known strain of Philippine native chicken that originated from Western Visayas. This native chicken is known as adapted to local agro climatic condition as well as for having the tasty meat, low cholesterol and packed with several nutrients and protein. Bamboo shoot on the other hand, is considered as a good source of fiber, potassium, antioxidants and lower in fat and calories. Capiz State University Dumarao Satellite College is fortunate enough to have these two research commodities, Darag Native Research Project which was funded by DOST PCAARRD and the Bamboo Research Project by the DENR Region 6. This two research project has already reached several milestones in its research endeavor such as Botanical Dewormer by Dr. Bede P. Ozaraga and is now on its commercialization phase. In order to further give innovations to these two commodities, several food products were developed such as DABAM food products. DABAM stand for DA – Darag Native Chicken and BAM – Bamboo shoot. These combinations of these two food products produces DABAM Siomai, Shanghai, and Crackers. To further enhance and level up this innovation, it is therefore, the purpose of this proposal to develop advanced food technology –ready-to-eat meal food products.			
(9-A) Project Description This study will be the development of several ready to eat food products of Darag native chicken and Bamboo shoot. Several food products will be developed such Darag native chicken in bottle/ pouch (Adobo, binakol, nilagpang, sisig, hamonado) and the combination of Darag native chicken and bamboo shoot such as DABAM burger patties. This will be a collaborative study between CapSU Dumarao researcher as the source of Darag native chicken and Bamboo shoots, Dr. Marife Hilapad, food expert from CapSU Mambusao and Prof. Jocelyn Legaspi, food expert from CApSU Dayao. Acceptability level will be taken first for this different food products to be conducted in the FFRDC, Dayao. This will followed by the evaluation of the shelf life and the nutritional value. Furthermore, the product from this study will also applied for utility Model certificate in the IPOPHIL and will be introduced in the market specifically in			

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the school canteen. The result of these study will be the answer for the on going DOST PCAARRD funded research on Native Chicken Value chain project where ready to eat food products from Darag native chicken will be developed and to be accessible in the market, ATBI and some souvenir shops in the region.

(9) Objectives

General Objectives:

Develop ready to eat Darag native chicken and Bamboo shoot food products.

Specific Objectives:

1. Evaluate the acceptability level of different Darag native chicken and Bamboo shoot ready to eat food products in terms of taste, texture, aroma, and color:
 - a. Darag Native Chicken and Bamboo Shoot Burger Patties
 - b. Darag Native Chicken Sisig
 - c. Darag Native Chicken Adobo
 - d. Darag Native Chicken Hamonado
 - e. Darag Native Chicken Binakol
2. Determine the type of packaging suited for the different food products.
3. Evaluate the shelf life and nutritional of different food product.
4. Find out the cost and return analysis of producing different ready to eat Darag native chicken and bamboo shoot food products.

(11) Review of Literature/Prior Art Search

Darag native chicken are indigenous strains which are located only in Western Visayas. (WESVARDEC,2009). They are well known for their adaptability to local agro climatic conditions,hardiness, ability to utilize farm by-products, tasty meat and a resistance to disease (PCAARD, 2007). One of the reasons that the native chicken are regaining their popularity in the Philippines is the fact that they have a unique, and arguably better, taste as compared to the commercial varieties. That applies to both their meat and their eggs (<http://poultrymanual.com>). Darag Chicken meat, when cooked, contains less fat, is high in protein, rich in potassium and has more unsaturated fatty acids.Famous local Darag delicacies are tinola, litson manok, tinu-om and chicken inasal (Fernandez, 2014).

Bamboo shoots on the other hand, is a good source of fiber, potassium, anti oxidants, and low in fats and calories. The juvenile shoots are not only delicious but are rich in nutrient components,mainly proteins, carbohydrates,minerals, and fiber and are low in fat and sugars. In addition, they contain phytosterols and a high amount of fiber that can be labeled as a nutraceuticals or natural medicine that are attracting the attention of health advocates and scientist alike. The shoots are free from residual toxicity and grow without the application of fertilizers (Chongtam *et al.*, 2011).

The following are the registered technologies taken from Google Patent (2022) search engine that are related to Darag native chicken and bamboo shoot food products:

- a. Darag native chicken and bamboo shoot burger patty.
GB2240252A – Preparation of Burger Patty by Patrick McDonagh from UK. He discloses the process for the preparation of a burger patty raw meat cuts comprising unfrozen meat cuts are ground and blended to a uniform temperature and composition. A sample of the meat is taken and analysed for fat content. The fat content of the meat is regulated by the further addition of fat or lean meat, if required, to maintain the fat content at an optimum value of between 16% and 18% by weight. After blending in seasoning ingredients, the meat is formed into patties, frozen and packed.
KR100716511B1 – Tuna Burger Patty and Tuna Burger comprising the same from South Korea. The inventor discloses a burger patty using tuna and burger comprising the same.
- b. Darag native chicken Sisig

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PHUM74672 – Canned fully cooked ready to eat rice mixed vegetable and chicken meat prepared in caldereta flavor by Mr. Ricardo S. Po.

- c. Darag native chicken Adobo
RU2510747C1 – Method of production or preserves “Chicken Stew”. The inventor discloses the method envisages recipe components preparation, chicken meat cutting and frying in melted fat, leek and white vegetables cutting and blanching, carrots blanching and cutting.
PHUM7490Z – Canned fully cooked ready to eat rice mixed vegetable and chicken meat prepared in adobo flavor by Mr. Ricardo S. Po.
- d. Darag native chicken Hamonado
PH2201800094Y1 - Tilapia (*Oreochromis niloticus*) fillet, hamonado variant and the process of making thereof. Inventors discloses tilapia fillet hamonado variant Is a convenient stable value - added food product characterized as having a light brown color with the preferred slight fishy odor and a palatable taste. The product was liked very much by the consumers.
- e. Darag native chicken Binakol
CN102524780A – Making Method of Can Native Chicken Soup with Chestnuts. The inventor discloses making method of a can of native chicken soup with chestnuts, which belongs to the technical field of food processing. The making method comprises the following steps: selecting a high-quality, healthy and disease-free native chicken, slaughtering, letting blood, scalding with hot water, removing hair, removing internal organs, washing and then cutting into blocks; further pre-boiling, stir-frying, adding chestnut kernels after removing shells and red skins, stewing with the native chicken twice at different temperatures for different time periods, canning, exhausting, sealing the can, performing high-temperature sterilization, cooling and performing other process steps.

This technologies, however, doesn't disclose the ingredients of the food products in this study.

(12) Scientific Basis/Theoretical Framework

Aliperio *et al.*, (2017) studies the acceptability level of bamboo shoots in making siomai and the result of their study showed that bamboo shoots can be used as substitute in making siomai. Furthermore, their study revealed that the quality of siomai has a significant differences in terms of taste, aroma and texture.

Barrios *et al.*, (2017) studies showed that 25% to 75% mixture of Darag native chicken and Bamboo shoot siomai was found satisfactory to very satisfactory acceptability level in terms of taste, aroma and texture.

(13) Methodology

Materials

The materials to be used in the study are chopping board, knife, pan, mixing bowl, mortar and pestle, wood ladle, whisk, packaging materials, cooking equipments and other necessary equipments for shelf life assessment and packaging. The ingredients will be the *Gigatochloa levis* (botong) shoots, Darag native chicken ground meat, ground pepper, seasonings, bread crumbs, garlic, onions, egg, water, cooking oil, cornstarch and all purpose flour and other specific ingredients for every food products.

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Methods

Cooking and Packaging

The materials to be used in this study will be taken from Darag native chicken Research Project and Bamboo Research Project from CapSU Dumarao. All the ingredients will also secured for every type of food products. After preparing the materials needed and ingredients, cooking and packaging will be done at FFRDC in Dayao, Roxas City to ensure the quality of the food products.

Sensory Evaluation of the Food Products

To evaluate the quality of the finished product sensory evaluation will be done using a 9 point hedonic scale in a qualitative description of Like Extremely (9), Like Very Much (8), Like Moderately (7), Like Slightly (6), Neither Like Nor Dislike (5), Dislike Slightly (4), Dislike moderately (3), Dislike very much (2) and Dislike Extremely (1). This will be done at FFRDC Laboratory. Semi trained panelist will be invited to taste the food products developed.

Determination of Shelf Life and Nutritional Value

To determine the shelf life of the product, samples will be taken and will be observed under room temperature and by using of a machine at CapSU Mambusao. To get nutritional value, samples will also be submitted in the DOST for laboratory analysis.

(14) Major Activities/Work plan

RESEARCH ACTIVITIES	Q1	Q2	Q3	Q4
Acquisition of materials				
Product Development and Analysis				
IP Protection Application				
IEC materials/Flyers production				
Technology Commercialization				
Extension of Technology to Community (Training)				

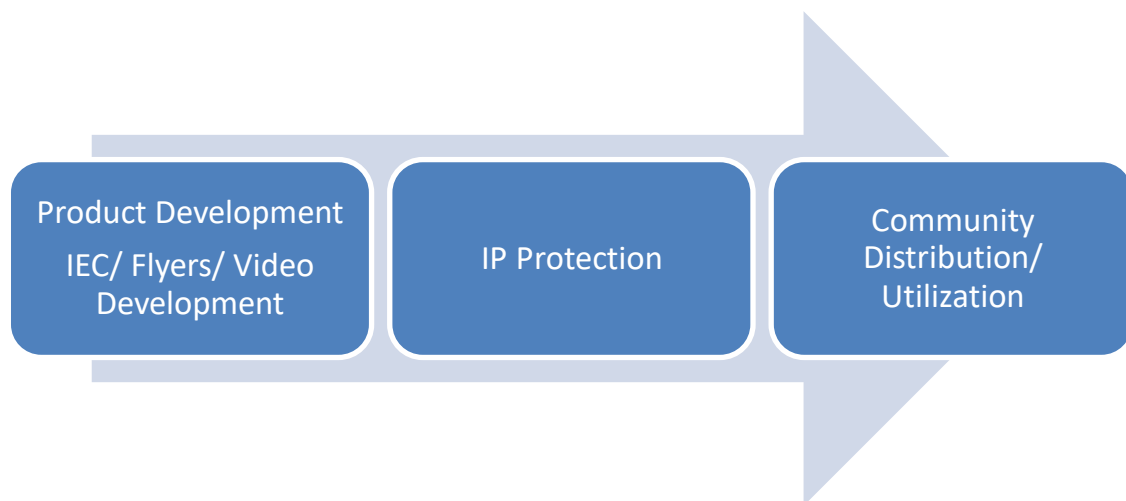
(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Train/ mentored native chicken growers group or association for the preparation of the ready to eat food products from Darag native chicken as well as for Bamboo. Help native chicken growers (CADAGA) for sustainable income in raising Darag native chicken. Help Capiz Toursim office for another food product that will be showcase in Pasalubong centers.	100
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials	5

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	for Darag native chicken and Bamboo shoot food products.		
Patents/IP	File 5 copyrights for the developed IEC materials and videos.	4	
Places and Partnerships	Forge linkages and partnerships to LGU's, Capiz Darag Growers Association, Capiz Tourism Office, DTI, and other organizations.	5	
Products	Produced ready to eat Darag native chicken adobo, hamondao, binakol, sisig and DABAM burger patties.		
Policy	Develop one policy inputs for the sustainable food production using Darag native chicken and bamboo shoot.	1	
Impact to Social/ Economics	<i>Social Impact:</i> • More available and accessible processed food products • Technology transfer and commercialization strengthened • Strengthened Public-Private Partnerships • Innovation ecosystem improved • Technopreneurship empowered • Local agribusinesses supported <i>Economic Impact:</i> • Income generated by Capiz Darag Growers Association Members and other native chicken growers.		

(15) Technology Transfer Activity Design



(16) Targets Beneficiaries

- Darag Native Chicken Raisers
- Bamboo Growers
- Entrepreneurs
- Nutritionists
- Parents
- Students

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(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Study Leader	Animal Science
Dr. Leah A. Ingalla	Co researcher	Animal Science
Dr. Marie Jean P. Penson	Co researcher	Education
Dr. Marife Hilapad	Co researcher	Food Tech
Prof. Jocelyn Legaspi	Co researcher	
Dr. Leo Andrew Biclar	Co researcher	
Dr. Efren L. Linan	Co researcher	

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (CapSU)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 @ 11,068 x 12 months	132, 816.00
B. Maintenance and Other Operating Expenses (MOOE)		
Transportation allowance		30,000
Representation Expenses		30,000
Training Expenses		50,000
Supplies (Cooking, Packaging, Office)		100,000
Communication		5,000
Others		20,000
CAPITAL OUTLAY		
Chicken Plucker Machine		100,000
Total		Php 335, 000.00

(19) Literature Cited

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